

General Comparison: Round Robin vs SJF

• Fairness versus efficiency

Round Robin focuses on fairness by giving each process an equal share of CPU time using time slices. This ensures that no process is starved and all processes make progress.

In contrast, SJF focuses on efficiency by always selecting the process with the shortest burst time. This minimizes average waiting and turnaround time but may cause long processes to wait for a long time.

• How Round Robin distributes CPU time

Round Robin distributes CPU time in a cyclic manner. Each process receives a fixed time quantum, and if it does not finish within that time, it is moved to the end of the ready queue.

This rotation ensures that all processes get CPU access regularly, making it suitable for systems that require responsiveness and fairness.

• How SJF Favors short jobs

SJF always selects the process with the smallest burst time among the available processes. As a result, short processes are executed quickly and experience very low waiting and turnaround times.

However, this behaviour can delay longer processes significantly, especially if short jobs keep arriving or if a long process starts first.

• Effect of the selected quantum on Round Robin behaviour

The time quantum plays a critical role in Round Robin performance.

- A small quantum increases responsiveness because processes are switched frequently, allowing faster initial execution for all processes.
- However, it also increases context switching overhead.
- A large quantum reduces context switching but makes Round Robin behave more like First-Come First-Served, reducing fairness and responsiveness.

Thus, choosing an appropriate quantum is essential for balancing performance and fairness.

• Effect on first response time and long-process treatment

Round Robin provides better first response time because every process gets CPU access early, even if only for a short duration. This makes it suitable for interactive systems.

SJF may delay the first response of some processes, especially if a long process starts first (as in non-preemptive SJF), causing other processes to wait.

For long processes:

- Round Robin allows them to progress gradually alongside other processes.
- SJF may delay them significantly, depending on arrival order and burst times.